Allan Wu

0413 360 799 | allan.wu@outlook.com.au | linkedin.com/in/allan-wu-engsci | Murdoch WA

SUMMARY

Engineering and computer science student with strong analytical and problem-solving skills, and experience in data analysis and visualisation. Passionate about applying technology to tackle problems in the resources and utilities sectors, particularly in automation and renewable energy.

TECHNICAL SKILLS

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- Programming languages and tools: Python, C, SQL, MATLAB, Java, JavaScript, Git/GitHub
- Electronic circuit simulation and design: LTspice, VHDL (ModelSim, Intel Quartus Prime)
- Data analysis and business automation: Microsoft Excel, Power BI, Power Apps, Power Automate

EDUCATION

Bachelor of Engineering (Honours) / Bachelor of Science **2023–Expected Dec 2027**
The University of Western Australia

- Combined degree majoring in Electrical & Electronic Engineering and Computer Science
- GPA: 6.893/7.000, WAM: 88.214%
- Faculty commendation for excellence: 96% in Computational Thinking with Python (2023)

Western Australian Certificate of Education **2017–2022**
Willetton Senior High School

- ATAR of 99.55
- Instrumental and Ensemble Music Performance program: first violin leader in string ensemble

COURSEWORK PROJECTS

Uni calendar web app (Agile Web Development) **Mar–May 2026**

- Effectively worked in a 4-member team to develop a Flask web application for tracking university schedules and sharing availability with friends, including reviewing frontend and backend code
- Led the design of the database and backend implementation, including server-side rendering with Jinja templates, routing, and end-to-end testing using Selenium

Group research report (Electronic Materials and Devices) **Mar–May 2025**

- Demonstrated technical writing and project management skills, collaborating in a 4-member team assignment on the environmental impacts of laptop computers and mitigation strategies
- Conducted literature reviews using academic databases, organised weekly meetings, and led final formatting and editing of the report for increased clarity and consistency

Source-to-source translator program (Systems Programming) **Aug–Sept 2024**

- Demonstrated proficiency in low-level programming with the design and implementation of a C11 program which translates, compiles and executes programs written in a small mini-language

WORK EXPERIENCE

Controls Engineer Intern **Nov 2025–Present**
Rio Tinto Iron Ore

- Performed Python data analysis of employee rosters to quantify annual hours spent on training
- Built automated reports and business applications to track workforce and mass transport fleet operations for Pilbara Projects (>1000 users), with software documentation for non-technical users
- Reviewed Manufacturer’s Data Reports, identifying discrepancies during rail project close-out

Waiter **Nov 2022–Oct 2025**
Winthrop Hawker’s Cafe

- Developed teamwork and communication skills, working effectively with other staff to maximise customer satisfaction: maintaining a clean and welcoming dining environment, greeting customers, serving orders and attending to customer requests in a fast-paced environment

English Tutor **Nov 2023–Jan 2024**
3E Learning

- Tutored primary and upper school students in English and Literature, developing lesson plans with tailored readings and assessments to improve students' creative and analytical writing skills